# Akshara Nadayanur Sathis Kanna

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## EDUCATION

**Carnegie Mellon University** Dec 2026
*Master of Science, Electrical and Computer Engineering (GPA: 3.9/4.0)* *Pittsburgh, PA*
- **Coursework:** Introduction to Machine Learning, Introduction to Deep Learning, Human-AI Interaction, Trustworthy AI, LLM Systems, AI Agents for Engineering, Designing & Prototyping AI Systems, Building Reliable Distributed Systems

**Shibaura Institute of Technology** Sep 2024
*Bachelor of Engineering (Honors), Computer Engineering (GPA: 3.9/4.0)* *Tokyo, Japan*
- **Achievements:** Presidential Award of Excellence - 2024

## SKILLS

**Programming Languages:** Python, C/C++, JavaScript/TypeScript, SQL, Bash, Go
**AI/ML:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, OpenCV, LangChain, Hugging Face Transformers, RAG
**Tools/Systems:** Jupyter, Git, AWS (EC2/S3), CUDA, Jetson Nano, Docker, Linux/WSL2, React, Node.js
**Evaluation:** Grader validation, Contamination audits, Terminal-Bench, ContextBench, DeepSWE, Red-teaming
**Spoken Language:** Japanese (JLPT N1)

## WORK EXPERIENCE

**AI Engineer Intern - LLM Evaluation & Benchmarking** **May 2026 - Aug 2026**
*Kanini Software Solutions Inc.* *Nashville, TN*
- Owned an **eleven-dimension benchmarking program** end-to-end for a ~25-binary Go LLM agent-orchestration platform.
- **Built evaluation infrastructure** with containerized eval harnesses and task suites over Terminal-Bench and SWE-Bench agent tasks with blinded LLM-judge panels, a token-matched RCT, and A/B experiments.
- Reached **97% retrieval recall** with a curated context pack 5x smaller than the auto-generated baseline; audited 2,931 production artifacts by sealed random sampling (N=75, 0 defects) and 216 prompt-injection trials at 0% attack success.

**Teaching Assistant for Introduction to Deep Learning (11-785)** **Nov 2025 - Present**
*Carnegie Mellon University* *Pittsburgh, PA*
- Debugged assignments covering **training loops, CNNs, sequence models, Transformers, and ResNets**, using PyTorch.

## RESEARCH EXPERIENCE

**LLM Evaluation and Multi-agent system Research** **May 2026 - Present**
*Shiba AI*
- Ran agent-panel experiments on **LLM collective consensus** (N=50, 340 trials, six providers) and a paired-name LLM hiring-bias audit.

**Explainable LLM-Guided Navigation with Orientation-Aware Planning (MiniGrid)** **June 2025 - Dec 2025**
*Learning & Control Group, Carnegie Mellon University*
- Built an **LLM–MPC MiniGrid controller** (GPT-4o + 50 sampled rollouts/step); solved obstacle navigation in 48 steps at 0.867 reward.

**Personalized Wearable Assistive Navigation for Visually Impaired Pedestrians** **Oct 2020 - Sep 2024**
*Honors Thesis, Mobile Multimedia Communications Laboratory, Shibaura Institute of Technology*
- Designed and prototyped a Jetson Nano–based wearable navigation system using **image processing**, **YOLO** object detection, and **depth estimation** to localize hazards and generate **personalized** step-based audio guidance; validated in controlled trials.

## PROJECTS

**Structure-Aware Cache Eviction for LLM Serving, CMU** **Jan 2026 - May 2026**
- Implemented **eviction policies** (weighted LFU, size-aware LRU) in **SGLang's RadixAttention prefix cache**; upstream PR.
- Cut mean TTFT 26% and p99 latency 39.6% vs. LRU across 2048 order-matched requests at equal throughput/evictions.

**AutoAnalyst: Multi-Agent Data Quality Debugger, CMU** **Jan 2026 - May 2026**
- Built an **MCP system** diagnosing ten data-defect classes, emitting executable fixes behind a **two-phase human-in-the-loop gate**.
- 100% detection recall across seven injected defect classes; fixes recovered 91% of downstream model degradation.

**Physics-informed Generative Modeling, CMU** **Aug 2025 - Dec 2025**
- Built next-frame prediction model for **PhysicsGen ball-bounce dynamics** using 50k+ paired Pymunk simulations in PyTorch.
- **Coordinate-conditioned cGAN improved localization;** ablations cut error by ~13% vs PhysicsGen baselines.

**BFine: Healthcare Recommendation App, Japan Venture Academy** **Sep 2024 - Dec 2024**
- Led a team building an AI-assisted clinic recommender for foreigners in Japan: user discovery, **matching and ranking pipeline**, API architecture design to support retrieval and explanation, and Figma UX; pitched at the **Google Tokyo JVA Expo**.

## PUBLICATIONS

- Collective Regimes in Multi-Agent LLMs under Reasoning Effort and Communication Topology
- Gender Bias in LLM Hiring Decisions: Evidence from Japanese Context and Mitigation Strategies Evaluation (ECML PKDD 2026)
- Personalized Wearable Assistive Navigation for Visually Impaired Pedestrians (WSSE 2024, **Best Paper Award)**